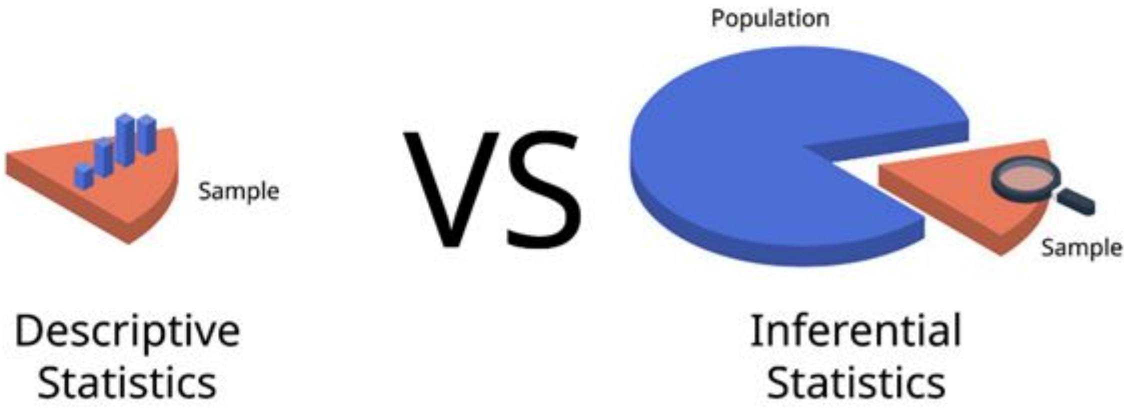
What is Statistics?

Statistics is a field of mathematics which focus on collecting, organizing, analyzing, interpreting, and presenting numerical data to understand complex information and make informed decisions.

Type of Statistics

The two main types of statistics are descriptive statistics and inferential statistics.



Descriptive statistics

Descriptive statistics summarize and organize data from a sample or a population. They provide a clear and concise overview of the data's main features without making any conclusions or predictions about a larger group.

- **Measures of Central Tendency:** The mean (average), median (middle value), and mode (most frequent value).
- **Measures of Dispersion:** The range (difference between the highest and lowest values), variance, and standard deviation, which show how spread out the data is.

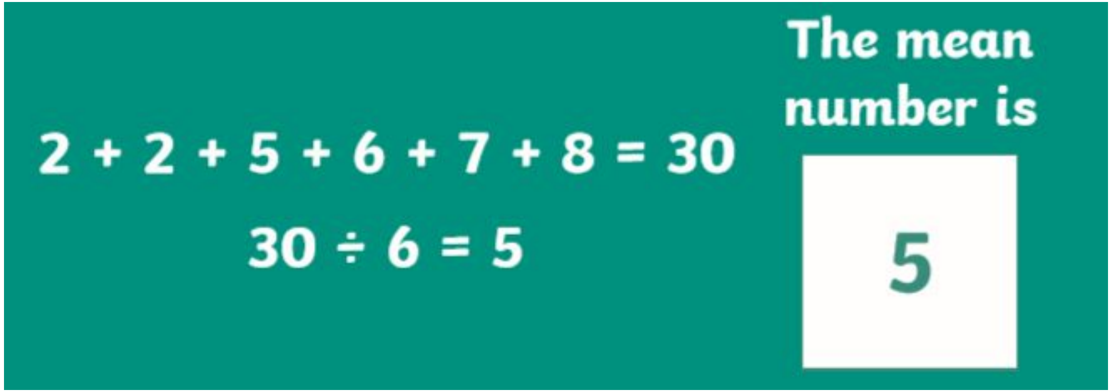
Measures of Central Tendency

These measures describe the center or typical value of a dataset. They tell you where most of the data points are clustered.

- **Mean:** The average of all the values.
- **Median:** The middle value when the data is arranged in order.
- **Mode:** The value that appears most frequently.

Mean

the arithmetic average, calculated by summing a set of numbers and dividing by the total count of numbers.



Median

The median is the middle value in a data set that has been arranged in order, meaning half the values are less than or equal to it, and half are greater than or equal to it. To find it, you first sort the numbers, and if there's an odd number of values, the median is the single middle number. If there's an even number, the median is the average of the two middle numbers.

1, 3, 3, **6**, 7, 8, 9

Median = 6

1, 2, 3, **4**, **5**, 6, 8, 9

$$\text{Median} = (4 + 5) \div 2$$

$$= 4.5$$

Outlier

An outlier is a data point that differs significantly from the rest of a dataset, lying far outside the overall pattern of the other observations. These extreme values, either much higher or lower than the rest, often indicate experimental errors, data entry mistakes, or genuine, rare anomalies in the data.

Income
54,000
35,000
5,80,000
53,000
68,000
49,000

Mean	1,39,833
Median	53,500

Mode

In statistics, the mode is the value that appears most frequently in a data set, representing the most common or popular observation.

Wheels	Product
2	Apple
4	Banana
2	Kiwi
2	Apple
8	Apple
2	Banana
2	Kiwi

Wheels	3.14
Wheels	2
Product	Apple

Measures of Dispersion (or Variability)

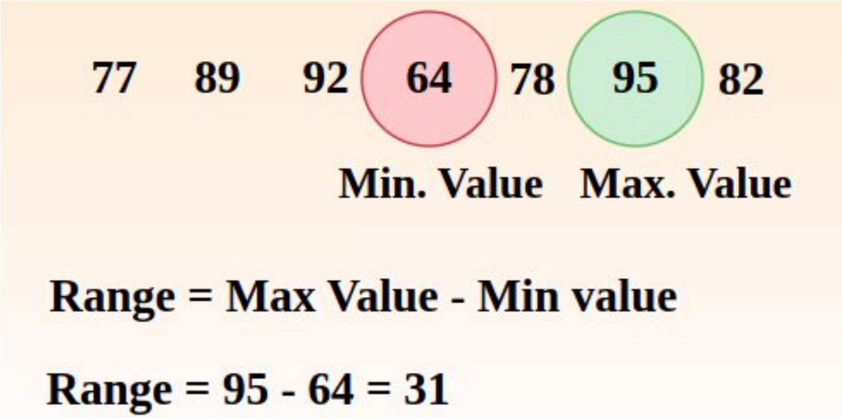
Measures of dispersion describe the spread or variability of data points within a dataset. Common measures of dispersion include the range, interquartile range, variance, standard deviation, and mean deviation. These non-negative values help determine how stretched or compressed a dataset is around its central tendency.

Common Measures of Dispersion

- **Range:** The simplest measure, it is the difference between the highest and lowest values in a dataset.
- **Interquartile Range (IQR):** The difference between the third quartile (75th percentile) and the first quartile (25th percentile), showing the spread of the middle 50% of the data.
- **Variance:** The average of the squared differences from the mean, indicating how far each data point is from the mean.
- **Standard Deviation:** The square root of the variance, providing a measure of dispersion in the original units of the data.
- **Mean Deviation:** The average of the absolute differences of each data point from the mean, showing the typical distance of data points from the mean.

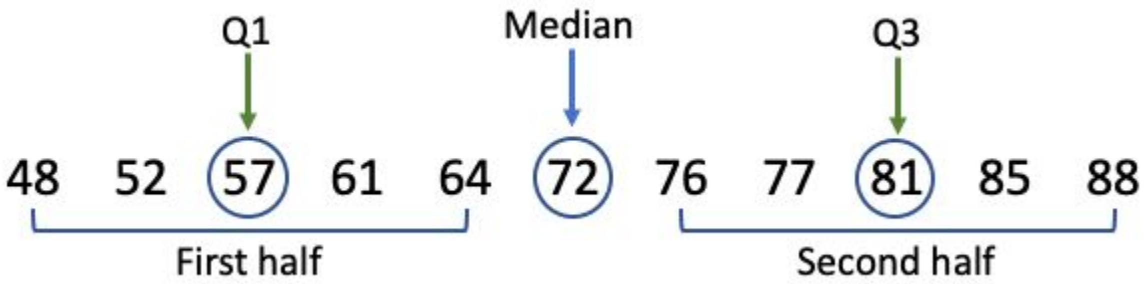
Range

range is a simple measure of data dispersion, calculated as the difference between the highest and lowest values in a data set (Range = Maximum Value – Minimum Value). It indicates the total spread of the data, providing a basic understanding of how far apart the data points are, though it can be misleading if outliers are present.



Interquartile Range (IQR)

Interquartile Range (IQR) is a measure of the spread or variability of the middle 50% of a data set, calculated as the difference between the third quartile (Q3) and the first quartile (Q1). The IQR is less sensitive to extreme values, or outliers, than the total range of the data.



Variance

Variance is a statistical measure of how spread out a set of data points is from their mean (average). A high variance indicates that data points are widely dispersed, while a low variance suggests they are tightly clustered around the mean.

$$\sigma^2 = \frac{\Sigma(xi - \bar{x})^2}{N}$$

Standard Deviation

A standard deviation is a statistical measure of the amount of variation or dispersion in a set of data points, essentially indicating how spread out the values are from the mean (average) of the dataset. A low standard deviation signifies that the data points tend to be clustered closely around the mean, while a high standard deviation indicates that the data points are more spread out over a wider range.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Mean Deviation

Mean Deviation (MD), also known as Mean Absolute Deviation (MAD), is a statistical measure of data dispersion that represents the average of the absolute differences between each data point

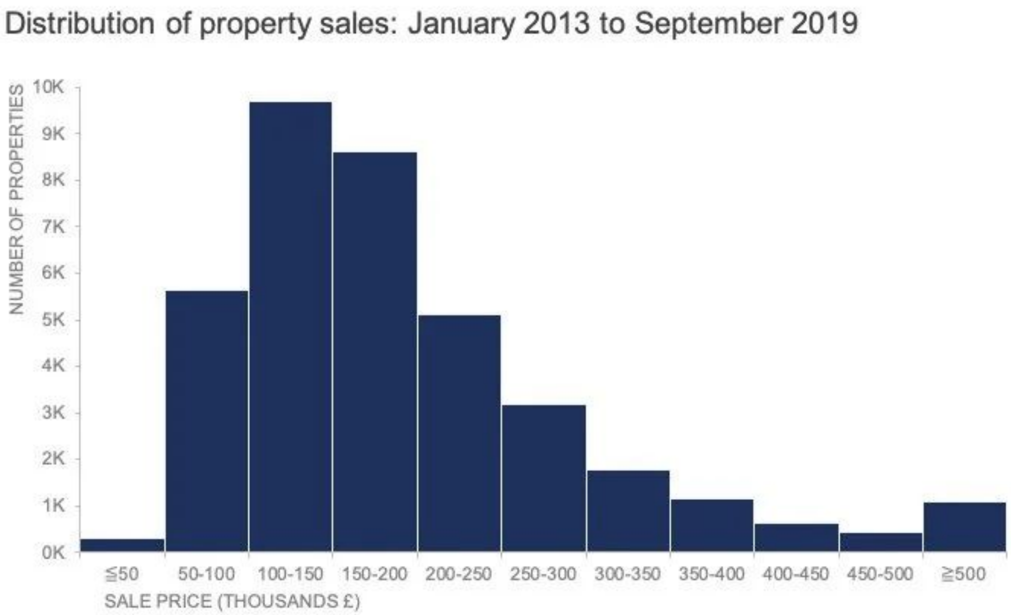
x_i	$\bar{x}$	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$
10	16	-6	6
12	16	-4	4
15	16	-1	1
19	16	3	3
24	16	8	8

$$MAD = \frac{\sum \lvert x_i - \bar{x} \rvert}{n}$$

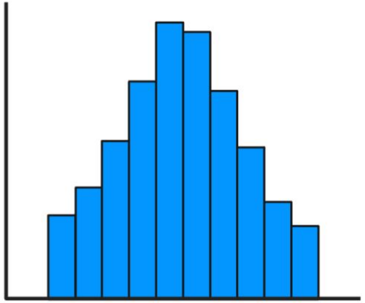
$$\bar{x} = \frac{\sum x_i}{n}$$

Histogram

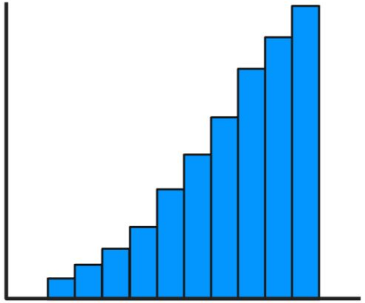
A histogram is a graphical representation of numerical data that displays the distribution of a dataset by showing the frequency of values within specified ranges called "bins".



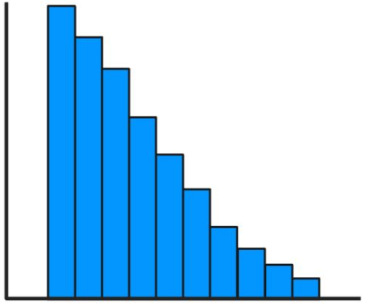
Histogram Distributions



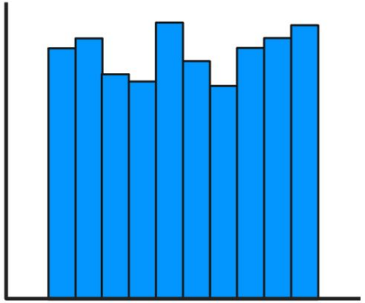
Normal distribution



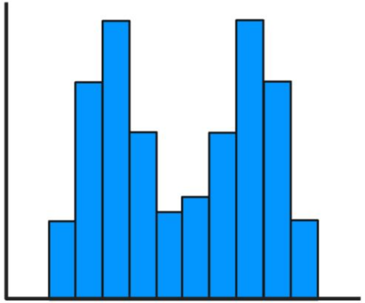
Left-skewed distribution



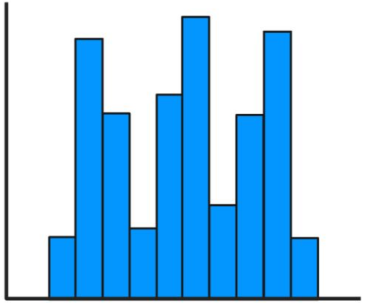
Right-skewed distribution



Uniform distribution



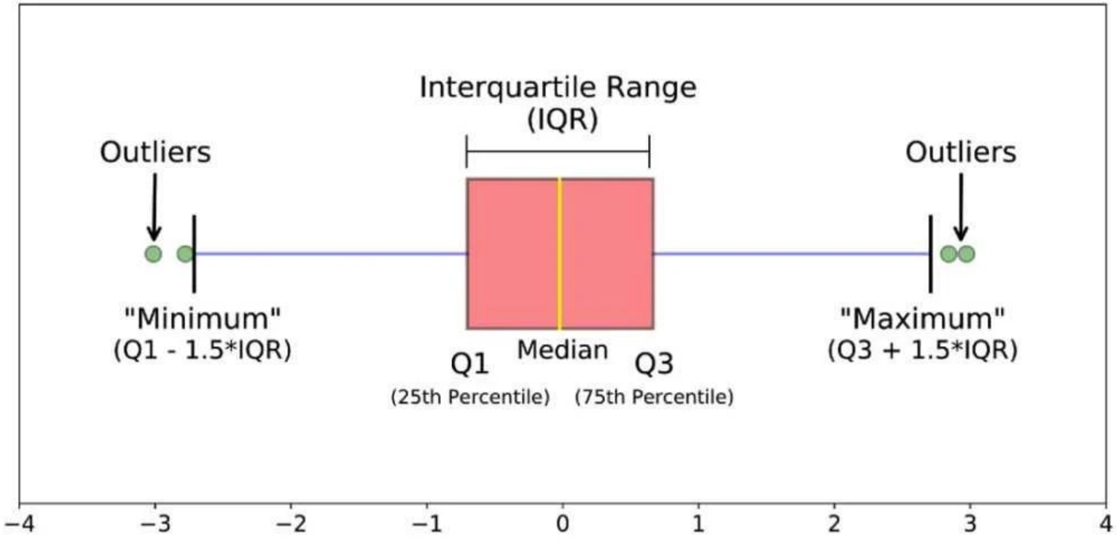
Bimodal distribution



Multimodal distribution

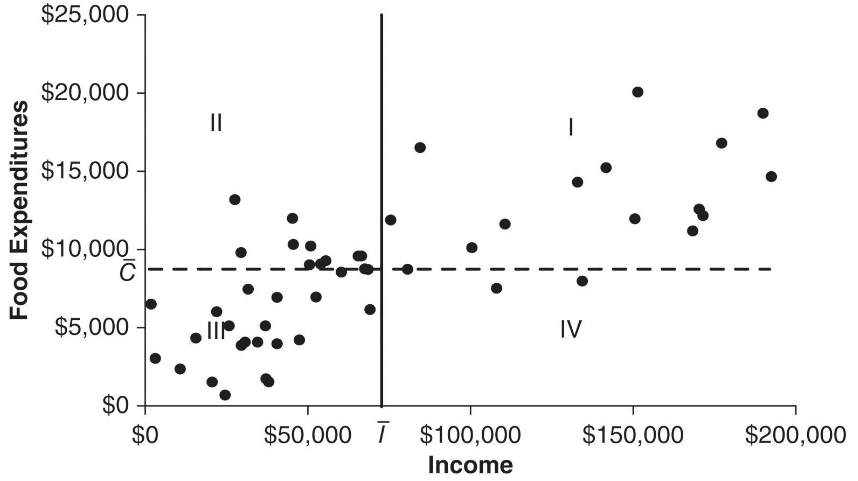
Box plot

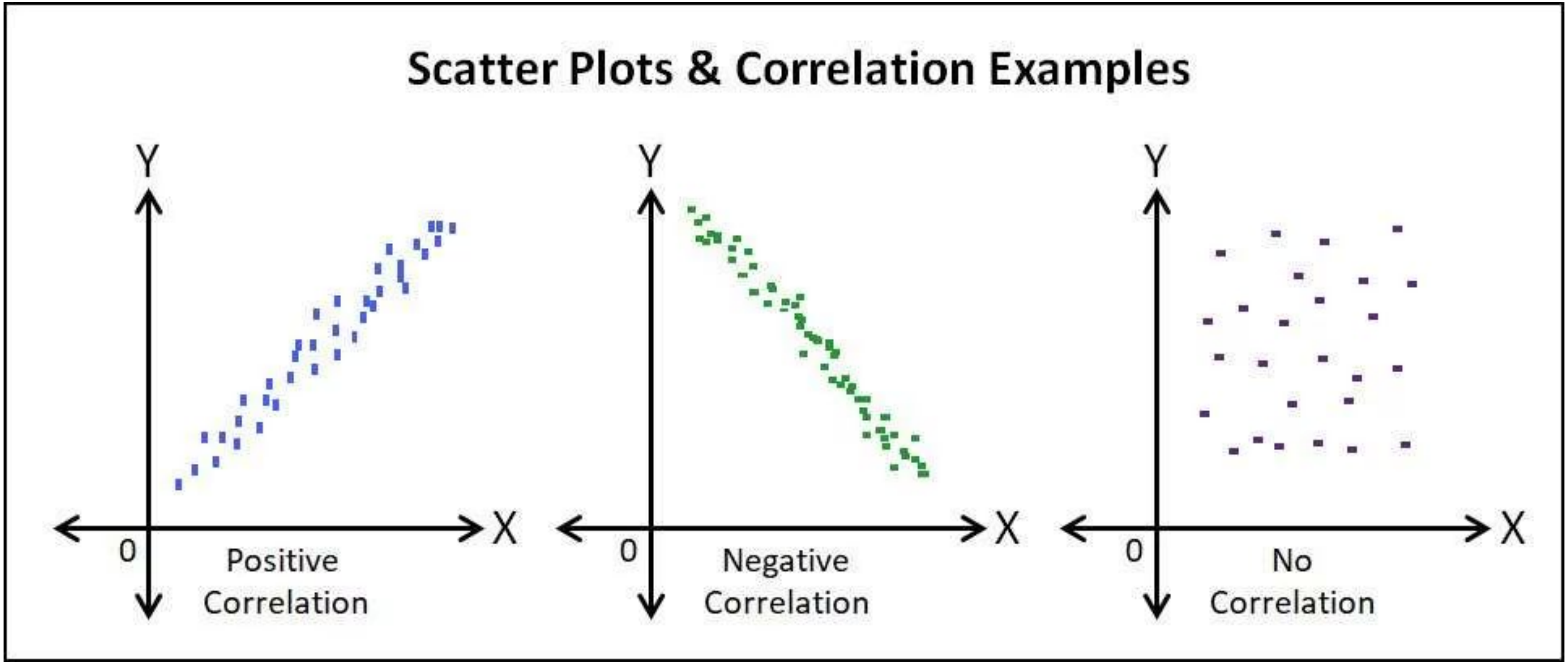
Box plot is a graphical representation of the distribution of a dataset. It displays key summary statistics such as the median, quartiles, and potential outliers in a concise and visual manner.



scatter plot

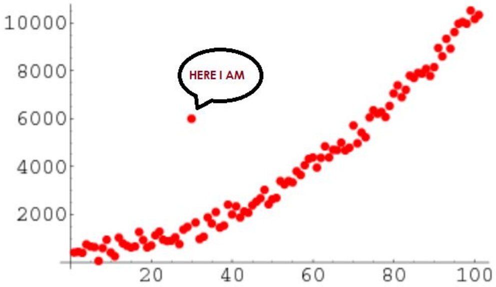
A scatter plot is a graph that visualizes the relationship between two numerical variables by displaying data points as dots on a two-dimensional coordinate plane. Each dot represents a single pair of data values, with its position determined by the corresponding values on the horizontal (x) and vertical (y) axes.

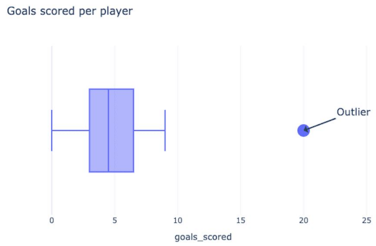


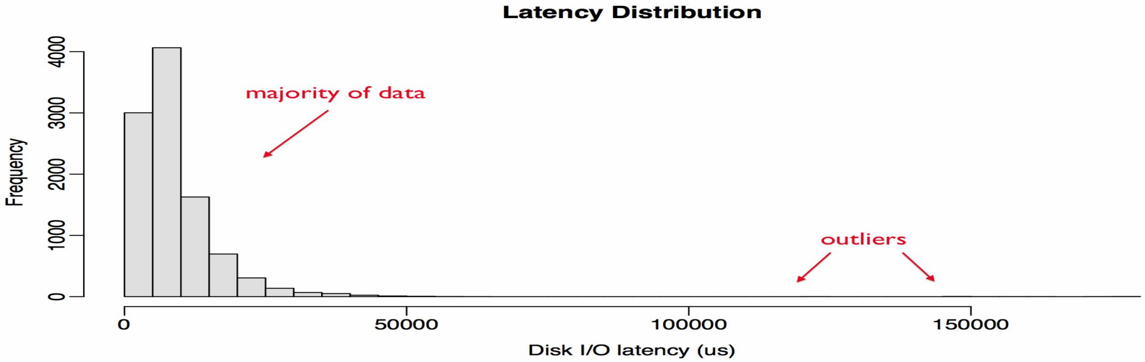

Scatter Plots & Correlation Examples
Y
X
0
Positive
Correlation
Y
X
0
Negative
Correlation
Y
X
0
No
Correlation

Outlier

An outlier is a data point that is significantly different from other observations in a dataset, lying far outside the overall pattern or range of the data







Inferential Statistics

Inferential statistics uses data from a random sample to draw conclusions and make generalizations about a larger population. It involves making predictions, testing hypotheses, and estimating population parameters, allowing researchers to understand trends and relationships beyond their specific sample data by using probability to assess the reliability of their findings.

- **Population and Sample**
- **Regression analysis**
- **Central limit theorem**